INCH-POUND

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Superseding
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MILITARY SPECIFICATION

RADIO FREQUENCY TRANSMISSION LINE ASSEMBLY CG-1013( )/U (Units of)

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the general requirements and tests for one type of surface wave transmission line assembly, the CG-1013( )/U.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command and Fort Monmouth, ATTN: AMSEL-ED-TO, Fort Monmouth, New Jersey 07703-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5820

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SPECIFICATIONS

FEDERAL

QQ-S-571 - Solder; Lead Alloy, Tin Lead Alloy, and Tin Alloy; Flux Cored Ribbon and Wire, and Solid Form.

MILITARY

MIL-M-13231 - Marking of Electronic Items
MIL-F-14072 - Finishes for Ground Signal Equipment
MIL-F-14256 - Flux, Soldering, Liquid Rosin Base

STANDARDS

MILITARY

MIL-STD-252 - Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

SIGNAL CORPS

SC-DL-99687 - Launching Unit
SC-DL-205925 - Case Transit
SC-DL-206324 - De-Icing Assembly

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

U.S. POST OFFICE DEPARTMENT

Parcel Post Regulations

(Applications for copies should be obtained from any first class U.S. Post Office).

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification

Consolidated Freight Classification Rule

(Application for copies should be addressed to the Consolidated Classification Committee 202 Union Station, Chicago 6, Illinois).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the National Classification Board 1424 Sixteenth St. NW, Washington 6, D.C.).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, the contractor shall furnish two sample units for first article inspection and approval (see 4.4 and 6.2).

3.2 Construction. The console shall be constructed in accordance with the following applicable drawings listed below.

SC-DL-99687	Launching Unit
SC-DL-205925	Case Transit
SC-DL-206324	De-Icing Assembly

3.3 Soldering.

3.3.1 Solder. Solder shall conform to Specification QQ-S-571. For soldering electrical connections, composition SN60, type AR or S shall be used for general purposes. When type S solder is used for soldering electrical connections, the flux shall conform to specification MIL-F-14256.

3.3.2 Acid or acid salts. No acid or acid salts shall be used in preparation for, or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical

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joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.3.3 Process. There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper following and wetting action, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.4 Cleaning.

3.4.1 Parts. After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted.

3.4.2 Units. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.5 Finish, protective. The unit shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware, such as hinges, screws and necessary touch up after mounting. The final paint film on type I surfaces shall be green color (olive drab), semigloss enamel matching a color chip provided by the procuring agency (see 6.7).

3.6 Marking of items. Marking shall conform to Specification MIL-M-13231 and shall be group I as described in that specification. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with a minimum disassembly of the items in which they are used.

3.6.1 Serial numbers. The De-Icing Assembly shall have serial numbers.

3.7 Tropicalization of material. Material shall be treated as follows:

3.7.1 Treating materials. Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.7.2 Toxicity. Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

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3.7.3 Flexibility. Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.7.4 Statement of treatment. The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use (see 6.2).

3.8 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment (see 4.8). Individual items shall not be handpicked for fit or performance; however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.9 Electrical characteristics. The Radio Frequency Transmission Line Assembly CG-1013( )/U shall meet the following electrical requirements (see 4.9).

3.9.1 Voltage-standing-wave-ratio. When tested as specified in 4.9.1, the maximum voltage-standing-wave-ratio (VSWR) shall not exceed 1.5 over the frequency range of 1700 mc to 2400 mc.

3.9.2 Insertion loss. The insertion loss of the transmission line assembly, when measured as specified in 4.9.2, shall not exceed 3.0 decibels (db).

3.9.3 High potential. After subjection to the high potential test of 4.9.3, the launcher housing shall show no visible deterioration and shall meet specified performance requirements.

3.9.4 De-icing assembly. When tested as specified in 4.9.4, the de-icing transformer secondary current shall be within the following limits.

Output terminal	Secondary current
10 volt	31 ±5 amps
14 volt	45 ±5 amps
19 volt	51 ±5 amps
24 volt	60 ±5 amps

3.10 Service conditions. The units shall meet the following service conditions:

3.10.1 Temperature.

a. Operating. Exposure in the range of +15° F to -40° F. (The +150° F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 24 hours, and at the low temperature extreme not to exceed 24 hours, at any one time (see 4.10).

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b. Nonoperating. Exposure in the range of +160° F to -80° F; exposure at each temperature extreme not to exceed 72 hours (see 4.10).

3.10.2 Relative humidity. Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation for 4 hours (see 4.10).

3.10.3 Rain. Two inches of rain fall per hour, with wind at 20 miles per hour (see 4.12).

3.10.4 Bounce and shock. The units shall meet the requirements of Table I.

3.10.5 Leakage. The launcher housings, when subjected to the tests of 4.15, shall meet the following requirements:

a. Pressure. Drop of pressure not to exceed 0.10 inches of mercury (Hg) in one minute.

b. Vacuum. Drop of pressure not to exceed 0.06 inches of Hg in five minutes.

TABLE I. Bounce and Shock.

Paragraph Heading	Paragraph No.	Required Performance After Test
Bounce	4.13	Full specification performance. No physical damage except surface abrasion.
Shock; drop	4.14	Full specification performance. Minor mechanical damage is permissible.

3.10.6 Preconditioning. The units shall meet the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.6 (also see 4.5).

3.11 Workmanship. The units on order shall be manufactured and assembled in accordance with the equipment drawings and the applicable portions of the following paragraphs herein:

3.2 Construction

3.3 Soldering

3.4 Cleaning

3.7 Tropicalization

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3.12 System Safety Engineering

3.12.1 Personal hazards. Personal hazards shall be kept to a minimum. The criteria of MIL-STD-454, requirements 1 and 8 shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual inspection (see 4.16).

3.12.2 Exposed edge rounding. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm), and exposed corners to a minimum of 0.5 inch (13 mm).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, or does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection of items before packing (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-810. The environmental tests shall be performed in the order shown in Table II according to the number of samples shown in the contract or order (see 6.2).

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4.4 First article inspection. This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4.1. and the inspections specified for group A, group B, and group C (see Tables III, V and VII respectively) and 4.5.

TABLE II. Order of Environmental Tests.

Test Heading	Requirement Paragraph	Inspection Paragraph
Non Damaging Tests		
Temperature	3.10.1	4.10
Moisture-resistance	3.10.2	4.11
Rain	3.10.3	4.12
Potentially Damaging Tests		
Bounce	3.10.4	4.13
Shock	3.10.4	4.14

4.4.1 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before packaging.

Item	Where required
Finish	3.5
Marking	3.6

4.5 Quality conformance inspection of items before packaging. The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4.1 and 4.5.1 through 4.5.3.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government--at its discretion--may perform all or any part of the inspection specified herein, to verify the contractor's compliance with specified requirements. (The amount of such verification inspection will depend on the extent to which the contractor's inspection procedures and records insure that the equipment on contract meets the specified procurement inspection (see 6.6). Each unit which will be subjected to group A, group B, or group C inspection shall be reconditioned after final assembly.

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4.5.1 Group A inspection. Units presented for acceptance shall be inspected for conformance with the criteria specified in Table III. Lots shall be formed and subjected to sampling in accordance with Table IV. Lots in which samples exhibit any failures shall be screened for that failure prior to being subjected to Group B inspection.

TABLE III. Group A Inspection.

Inspection Heading	Requirement Paragraph	Inspection Paragraph
Visual and mechanical	3.9	4.7
Electrical		
Voltage-standing-Wave-Ratio	3.9.1	4.9.1
High Potential	3.9.3	4.9.3
De-Icing Assembly	3.9.4	4.9.4

TABLE IV. Sampling plan for group A inspections.

Lot Size	Sample Size
2 to 8	*
9 to 15	13
16 to 25	13
26 to 50	13
51 to 90	13
91 to 150	13
151 to 280	20
281 to 500	29
501 to 1200	34
1201 to 3200	42
3201 to 10,000	50
10,001 to 35,000	60
35,001 to 150,000	74
150,001 to 500,000	90
500,001 and over	102

1. * Indicates entire lot must be inspected.
2. Acceptance number in all cases is zero.

4.5.2. Group B inspection. Group B inspection shall be performed on lots that have passed Group A inspection. This inspection shall consist of the inspections listed in Table V. Each lot shall be subjected to sampling inspection in accordance with Table VI. Lots in which samples exhibit any failures shall be screened for that failure prior to units within the lot being subjected to Group C inspection.

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4.5.2.1 Order of inspection within group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

TABLE V. Group B Inspection.

Inspection Heading	Requirement Paragraph	Inspection Paragraph
Interchangeability	3.8	4.8
Insertion Loss	3.9.2	4.9.2

TABLE VI. Sampling plan for group B inspection.

Lot Size	Sample Size
2 to 8	2
9 to 15	2
16 to 25	3
26 to 50	5
51 to 90	5
91 to 150	6
151 to 280	7
281 to 500	9
501 to 1200	11
1201 to 3200	13
3201 and over	15

1. The acceptance number in all cases is zero.

4.5.2.2 Group C inspection. This inspection shall be as listed in Table VII and shall be performed on sample units that have been subjected to and met group A and group B inspection.

TABLE VII. Group C Inspection.

Inspection Heading	Requirement Paragraph	Inspection Paragraph
Leakage Test	3.10.5	4.15

4.5.3 Sampling for inspection of equipment. Two launching units for the group C inspection shall be selected each month, without regard to their quality, except that the units inspected at the start of the contract shall be selected from the first units produced.

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4.5.3.1 Noncompliance. If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which are manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.2).

4.5.3.2 Reinspection of conforming group B and group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.6 Bounce preconditioning . The unit shall be placed in its normal operation position on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning , the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.7 Visual and mechanical inspection. The units shall be examined for the defects listed in MIL-STD-252.

4.8 Inspection for interchangeabil ity. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeabil ity requirements of 3.8. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. Location of fasteners on case and cover of the de-icing transformer.

b. Location of hinges and fasteners on the transit case and cover.

c. Location of fasteners, mounting-holes, and hold-down straps for securing parts within the transit case.

4.9 Electrical tests. The transmission line assembly specified in 4.9.1 and 4.9.2, shall consist of 140 feet of the dielectric coated transmission line terminated by two launcher units. The line shall be tensioned to approximately 100 pounds.

4.9.1 Voltage-standing-wave-ratio. The VSWR shall be determined by connecting a signal generator to the input through a slotted line and terminating the output of the assembly in a load of 50 ohms impedance having a VSWR of 1.1 or less, Termaline coaxial load resistor , Model 80-5, or equal. The frequency of the signal generator shall be varied from 1700 MHz to 2400 MHz while moving the detecting probe on the slotted line in order to observe the maximum

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standing wave ratio on a millivoltmeter or equivalent indicator, which shall be as specified in 3.9.1.

4.9.2 Insertion loss. Insertion loss shall be determined by terminating the transmission line assembly in a calibrated bolometer or crystal mount having an impedance of 50 ohms. The insertion loss shall be measured at every 100 MHz within the frequency range of 1700 MHz to 2400 MHz by first establishing reference levels on the output indicator directly connected to the signal generator. The assembly under test shall then be inserted between the calibrated output indicator and the signal generator. The generator power output shall be increased until the original reference levels are obtained. The power output increase, expressed in decibels, shall comply with the requirements of 3.9.2.

4.9.3 High potential. The high potential test performed on the launcher housing shall consist of one minute applications of 500 volts, 60 hertz, between the coaxial fitting and the metal housing and between the housing and the mounting.

4.9.4 De-Icing assembly. The de-icing transformer shall be tested using the following procedure:

a. The load shall consist of two launcher units, 40 feet of 8 AWG wire, and 2.43 feet of .081 inch diameter nichrome wire, which gives the equivalent resistance of 250 feet of transmission line.

b. Place the load across the 14 volt transformer terminal and turn the transformer "ON". Adjust the primary voltage to 115 volts, and record the secondary current.

c. Repeat (b) above for the 10 volt, 19 volt, and 24 volt outputs except that the primary voltage shall not be readjusted. The secondary currents shall be as specified in 3.9.4.

4.10 Temperature test. The transmission line assembly shall be subjected to the temperature cycle shown on Standard MIL-STD-810 and shall meet full specification requirements at steps 1, 4, 5, 8, and 10.

4.11 Moisture resistance test. The transmission line assembly shall be tested as follows:

4.11.1 Test conditions.

a. Do not remove the de-icing transformer from the test chamber for measurements. The launching units may be removed; however, tests shall be made immediately upon removal.

b. Start measurements not more than 5 minutes after power is applied. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.11.2 Test methods.

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a. Dry at 130° F ±5° F, for 24 hours.

b. Condition at 77° F ±5° F and 40 to 50 percent relative humidity for 24 hours.

c. Take measurements as specified in 4.11.3, and make adjustments as necessary to meet full specification requirements.

d. Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to standard MIL-STD-810. The measurements specified in 4.11.3 shall be made during the periods shown on the standard and shall comply with 4.11.3.

e. After cycling has been completed, condition the units for 24 hours at 77° F and 40 to 60 percent relative humidity. The units shall meet full specified performance for the measurements specified in 4.11.3.

4.11.3 Performance. The units under test shall meet full specification performance during cycling. There shall be no mechanical deterioration, as evidenced by corrosion of metal parts or binding or rotating parts, and there shall be no degradation in performance from that obtained prior to the 5 humidity cycles of 4.11.2(d).

4.11.4 Failure. If any unit fails to meet the performance specified in 4.11.3 during cycling or after conditioning as specified in 4.11.2(e), it does not pass the test.

4.12 Rain test. Each transit case and transformer shall be tested step by step as follows: a. Dry at 150° ±5° F for 48 hours.

b. Condition for 4 hours at 77° ±5° F and 40 to 60 percent relative humidity.

c. Place the unit in its normal upright position, in a test chamber maintained at 77° ±5° F. Allow water at 65° ±10° F to fall freely on all exposed surfaces of the unit, at a rate of 2 inches of rainfall per hour. Maintain the condition of vertical, simulated rainfall for 2 hours.

d. With the rainfall continuing at the 2-inch rate, introduce a simulated wind blowing horizontally at 20 ±5 miles per hour against one side of the unit. The water shall fall on all exposed surfaces and be blown by the wind against the side of the unit. Maintain this condition for 1 hour.

e. Rotate the unit 90 degrees horizontally, to expose, a second side of the unit to the wind and rain. Maintain this condition for 1 hour.

f. Continue as in 4.12(e) until all four sides of the unit have been exposed to the wind and rain, each side for 1 hour.

g. Open the unit; there shall be no visible evidence of leakage. (Condensation of moisture inside of the unit may occur but its pattern is distinguishable from that of water resulting from leakage.)

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4.13 Bounce test. The transmission line assembly in its transit case and the de-icing transformer shall be tested on the package tester, as made by the L.A.B. Corporation, Skaneateles, New York, or equal, as follows:

a. Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

b. Place the unit, prepared for field transport, on the bed of the package tester, limiting the lateral motion (by wooden fences) to not more than 3 inches and not less than 1 inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the unit against the back stop is not restrained.

c. Operate the package tester, shafts in phase, for a total of 3 hours at 284 ±2 rpm. Turn the unit at the end of each 30 minutes so it will rest on a new face.

4.14. Shock test; drop. The transmission line assembly in its transit case shall be dropped a distance of 2 feet on each corner, face, and edge for a total of 26 drops. Drops shall be made from a quick-release hook or pendulum tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The floor or barrier receiving the impact shall be of 2 inch fir backed by concrete or a rigid steel frame.

4.15 Leakage test. The launcher housing under test shall be connected to a pressure supply line from a compressor or vacuum pump. A needle valve shall be placed in the supply line, to isolate the launcher housing pressure from the supply source; and a mercury monometer shall be placed between the needle valve and the housing, to measure the pressure differential in the unit above or below atmospheric where atmospheric pressure is 29.92 inches of mercury.

a. Pressure. Apply pressure to the unit, by use of a compressor, until an equilibrium is reached and a differential of 5 psi (10.13 inches of Hg) above atmospheric is observed on the monometer. When this positive pressure is applied, the pressure drop over a one minute period shall be within the limit specified in 3.10.5(a).

b. Vacuum. Apply a vacuum to the unit, by use of a vacuum pump, until an equilibrium is reached and a differential of 3 psi (6.12 inches of Hg) below atmospheric is observed on the monometer. When this negative pressure is applied, the pressure drop over a five minute period shall be within the limit specified in 3.10.5(b).

4.16 System Safety Inspection. An inspection shall be performed to verify compliance with those portions of 3.12 which can be determined visually.

5. PACKING

5.1 Packing requirements. The packing requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.2).

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6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Radio Frequency Transmission Line Assembly CG-1013( )/U is intended to be used as part of the Radio Set AN/TRC-29( ) series used in the rear-area radio relay system.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

c. Type required.

d. First article inspection.

e. Marking and shipping of samples

f. Location of final inspection

g. Technical literature required.

h Quantity of tools and running spare parts required.

i. Submission of the statement of treatment referenced in 3.7.4, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example CG-1013W/U. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.4 Location of leakage test. It is desirable that the leakage test (4.15) be performed at a location that will minimize handling (which might cause damage to the units) after this inspection is completed. It is recommended that the entire lot (including all previously inspected samples units) be sampled and inspected immediately prior to packaging.

6.5 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.2.2).

6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of

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verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality history of the product (see 4.5.1, 4.5.2, and 4.5.3).

6.7 Color. The color chip furnished by the procuring agency will be color chip No. X24087.

6.8 Subject term (key word) listing.

Radio Set AN/TRC-29( )
Rear-area radio relay system
Launching unit
Case transmit
De-icing assembly

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-CR

Preparing activity:
Army-CR
(Project 5820-A880)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-12170B (CR)	90/09/18

3. DOCUMENT TITLE: RADIO FREQUENCY TRANSMISSION LINE ASSEMBLY CG-1013()/U (Units of)

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Commander, U.S. Army Communications Electronics Command	(1) Commercial (908) 532-5851	(2) AUTOVON 992-5851
c. ADDRESS (Include Zip Code) ATTN: AMSEL-ED-TO (Stdzn Ofc) Ft. Monmouth, NJ 07703-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340. AUTOVON 289-2340	

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